Digital Forensics 1

Rubric 1
Forensic Investigation Process

The project must include complete documentation of a forensic investigation process following proper methodology. Apply evidence handling procedures maintaining data integrity during collection. Create a chain of custody log with timestamps and handler identification. Implement initial response techniques to secure digital evidence. Documentation must include all steps taken with justification for methodologies used.

Step 1: Extract Your Strict Technical Requirements

Non-negotiable items I must submit.

- **Initial Response:** List of the exact commands or steps I will take to isolate your target machine (e.g., disconnecting a VM network adapter, dumping RAM).

  Dumping RAM

The error `No such file or directory` pointing to `/lib/modules/6.19.14+kali-amd64/build` confirms that your system is missing the required **Linux Kernel Headers** for your current operating system version. Without these files, `make` cannot compile the module.

To fix this, install the missing headers and clean your folder by running these commands:

1. Install the Kernel Headers

Run this command to download the precise package matching your current `6.19.14+kali-amd64` kernel:

bash

```
sudo apt update && sudo apt install linux-headers-$(uname -r) -y
```

Use code with caution.

2. Clean and Recompile

Once that installation completes, clean out the broken configuration records and run the build utility again:

bash

```
make clean
```

```
make
```

Use code with caution.

After `make` finishes, your file will compile successfully. You can verify it exists by running `ls -l *.ko`.

- **Evidence Handling:** Define your collection tools (e.g., FTK Imager, `dd`) and how you will document write-blocking.
- **Data Integrity:** Note exactly where and when you will generate cryptographic hashes (MD5/SHA-256).

- **Chain of Custody:** Determine the exact metadata required (Timestamp, Handler Name, Item ID, Transfer Reason).

Step 2: Structure Your Personal Grading Matrix

Since you are grading your own progress, set up a **3-Tiered Self-Evaluation System** to prevent easy mistakes:

- **Tier 3 (Done right):** The evidence is uncompromised, hashes match, and the "why" is fully explained.
- **Tier 2 (Done, but risky):** The step was completed, but you forgot to explain your methodology or missed a timestamp.
- **Tier 1 (Failed):** The action altered file metadata, or a timeline gap exists.

Step 3: Draft Your Evidence Verification Checklist

Turn the rubric into an actionable checklist that you run through *before* submitting. Group them by phase:

- **Phase 1: Phase Preservation.** Did I document the live state before pulling the plug?
- **Phase 2: Acquisition Verification.** Do I have matching pre-imaging and post-imaging hashes clearly written down?
- **Phase 3: The Paper Trail.** Does my Chain of Custody log have a flawless, gap-free timeline from the moment I "found" the evidence to submission?

Step 4: Add the "Justification" Layer

The prompt specifies you must include *justification for methodologies used*. For every tool you run, your rubric needs a rule verifying you answered:

- *Why* did you choose this tool over another?
- *How* does this specific tool protect data integrity?
